Exercise Sheet 6

Throughout, E_{jk} is the matrix unit with 1 in row j and column k. For $\sigma \in S_n$ the permutation matrix P_σ is defined by $P_\sigma e_j = e_{\sigma(j)}$. Exercises 6.1–6.5 are the core work. Exercise 6.6 is an optional extension.

Exercise 6.1. Let $n \geq 2$ and $\mathfrak{u}(n) = \{X \in M_n(\mathbb{C}) : X^* = -X\}$. Put $\langle X, Y \rangle = -\operatorname{tr}(XY)$.

(a) Show that $\langle \cdot, \cdot \rangle$ is a real inner product on $\mathfrak{u}(n)$. Show that $\langle gXg^{-1}, gYg^{-1} \rangle = \langle X, Y \rangle$ for $g \in \mathrm{U}(n)$.

(b) Show that $\mathfrak{z}(\mathfrak{u}(n)) = i\mathbb{R}I$.

(c) Use the splitting proposition of this lecture to show $[\mathfrak{u}(n), \mathfrak{u}(n)] = \mathfrak{su}(n)$.

(d) Let $n = 2$ and

$$X_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad X_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Find the matrix of $\operatorname{ad} X_1$ on $\mathfrak{su}(2)$ in the basis X_1, X_2, X_3. Check that it is skew-symmetric for $\langle \cdot, \cdot \rangle$. Compute the Killing form value $B(X_1, X_1)$.

Exercise 6.2. Let $K = \mathrm{SO}(3)$. Take as given that $\mathrm{SO}(3)$ is connected. The torus theorem needs this hypothesis. For $\theta \in \mathbb{R}$ put

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad T = \{R_\theta : \theta \in \mathbb{R}\}, \qquad w = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(a) Let $\theta \notin \pi\mathbb{Z}$. Show that every $g \in \mathrm{SO}(3)$ commuting with R_θ lies in T. Deduce that T is a maximal torus.

(b) Show that $w \in \mathrm{SO}(3)$ and $wR_\theta w^{-1} = R_{-\theta}$.

(c) Show that $N_K(T) = T \cup wT$, so $W(K, T) \cong \mathbb{Z}/2$.

(d) Use the torus theorem to show that every element of $\mathrm{SO}(3)$ is a rotation about some axis. Show that rotations by θ and $-\theta$ about one axis are conjugate.

Exercise 6.3. Let $T \subseteq \mathrm{U}(3)$ be the diagonal unitary matrices, with Lie algebra $\mathfrak{t} = \{i \operatorname{diag}(a_1, a_2, a_3) : a_j \in \mathbb{R}\}$. Let $T_1 = T \cap \mathrm{SU}(3)$ and $\omega = e^{2\pi i/3}$.

(a) Write down $P_{(12)}$ and $P_{(123)}$. Compute $P_\sigma \operatorname{diag}(z_1, z_2, z_3) P_\sigma^{-1}$ for both by matrix multiplication.

(b) Which permutation matrices lie in $\mathrm{SU}(3)$? For each transposition τ give an element of $\mathrm{SU}(3)$ that represents the class of τ in $W(\mathrm{SU}(3), T_1)$.

(c) For $H = i \operatorname{diag}(a_1, a_2, a_3)$ we have $[H, E_{jk}] = i(a_j - a_k)E_{jk}$. Let ε_j be the $\mathbb{C}$-linear form on $\mathfrak{t}_\mathbb{C}$ with $\varepsilon_j\big(i \operatorname{diag}(a_1, a_2, a_3)\big) = a_j$. List the roots of $\mathfrak{u}(3)$ in terms of the ε_j. Compute the reflection s_α for $\alpha = i(\varepsilon_1 - \varepsilon_2)$, using the inner product $-\operatorname{tr}(XY)$. Compare with (a).

(d) Show that $\operatorname{diag}(1, \omega, \omega^2)$ and $\operatorname{diag}(\omega, 1, \omega^2)$ are conjugate in $\mathrm{SU}(3)$. Give an explicit conjugating element and check by multiplication.

Exercise 6.4. This exercise contrasts $\mathrm{U}(2)$ and $\mathrm{SU}(2)$ with $\mathrm{SL}(2,\mathbb{R})$.

(a) Let
$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \mathrm{U}(2).$$
Find $X \in \mathfrak{u}(2)$ with $\exp X = U$ by diagonalising U.

(b) Let
$$V = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SU}(2).$$
Find $X \in \mathfrak{su}(2)$ with $\exp X = V$.

(c) Let
$$g = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}).$$
Compute g^m for $m \in \mathbb{Z}$. Show that g lies in no compact subgroup of $\mathrm{SL}(2,\mathbb{R})$.

(d) For $Y \in \mathfrak{sl}(2,\mathbb{R})$ show that $\operatorname{tr}\exp Y \geq -2$, with equality only if $\exp Y = -I$. Deduce that $g \notin \exp(\mathfrak{sl}(2,\mathbb{R}))$.

(e) Let $H = \operatorname{diag}(1,-1)$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Show that $\mathbb{R}H$ and $\mathfrak{so}(2) = \mathbb{R}J$ are maximal abelian subalgebras of $\mathfrak{sl}(2,\mathbb{R})$ that are not Ad-conjugate. Show that $\mathfrak{sl}(2,\mathbb{R})$ has no Ad-invariant inner product.

Exercise 6.5. Define $\mu\colon \mathrm{U}(1)\times\mathrm{SU}(2) \to \mathrm{U}(2)$ by $\mu(z,A) = zA$. Put
$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(a) Show that μ is a surjective homomorphism.

(b) Show that $\ker\mu = \{(1,I),(-1,-I)\}$ and that it is central.

(c) Show that $Z(\mathrm{U}(2)) = \{zI : |z| = 1\}$ and $Z(\mathrm{SU}(2)) = \{\pm I\}$. Identify T_0, K_{sc} and F of the structure theorem for $K = \mathrm{U}(2)$. You may use that $\mathrm{SU}(2)$ is simply connected.

Exercise 6.6 (Optional extension). Let $K = \mathrm{U}(2)$ with diagonal maximal torus T, and let $t = \operatorname{diag}(e^{i\theta_1}, e^{i\theta_2})$. Put $\varphi = \theta_1 - \theta_2$. Let
$$Y_1 = E_{12} - E_{21} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Y_2 = i(E_{12} + E_{21}) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},$$
a basis of $\mathfrak{t}^{\perp}$ in $\mathfrak{u}(2)$ for the form $-\operatorname{tr}(XY)$.

(a) Compute $\operatorname{Ad}(t^{-1})Y_1$ and $\operatorname{Ad}(t^{-1})Y_2$. Write the matrix of $\operatorname{Ad}(t^{-1})$ on $\mathfrak{t}^{\perp}$.

(b) Show that $\det\big(\operatorname{Ad}(t^{-1}) - I\big)|_{\mathfrak{t}^{\perp}} = |e^{i\varphi} - 1|^2 = 4\sin^2(\varphi/2)$.

(c) For which t does this determinant vanish? Relate the answer to the preimages of the conjugation map Φ from the lecture.